NeuroPpred-PSCG: A multimodal framework using ProtT5 and structural features for neuropeptide prediction based on gated cross-attention mechanism and multi-scale gated convolution

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Abstract. As fundamental signaling molecules, neuropeptides are involved in diverse physiological regulatory activities. Abnormal expression patterns have been closely linked to neurological and metabolic disorders, which underscores the urgent need for precise and rapid identification methods. Here, we develop NeuroPpred-PSCG, a novel multimodal deep learning framework. At its core, the framework employs both a bidirectional cross-attention mechanism and a gated fusion module to integrate global semantic insights from the protein language model (ProtT5) with local conformational data derived from secondary structure. Furthermore, it utilizes a multi-scale gated convolutional network to enhance hierarchical contextual information. On the independent test set, NeuroPpred-PSCG surpasses current leading methods on most metrics, achieving ACC of 94.8%, SN of 95.1%, F1-score of 94.8%, and MCC of 0.896, reflecting robust overall discriminative capability. Compared to the previous best-performing model, NeuroPpred-MSN, our method shows improvements of 1.2% and 0.024 in ACC and MCC, respectively, and most notably, a significant 2.8% gain in SN. Ablation studies, robustness analysis, and interpretability analysis further substantiate the effectiveness of each component, as well as the overall stability and interpretability of NeuroPpred-PSCG. The source code and datasets have been made publicly accessible at https://github.com/xinyizhang186/NeuroPpred-PSCG.

Keywords: Neuropeptide; Protein language model; Gated cross-attention mechanism; Multi-scale gated convolution network; Interpretability analysis

1 INTRODUCTION

Neuropeptides are small, bioactive molecules secreted by the nervous system, typically ranging from 3 to 100 amino acids in length [1, 2]. As critical signaling mediators, neuropeptides are involved in a wide array of physiological processes, including learning and memory, mood regulation, sleep-wake cycles, and energy metabolism [3, 4]. Neuropeptide dysregulation is implicated in a spectrum of neurological and metabolic disorders, such as Alzheimer's disease, Parkinson's disease, depression, and obesity [5-7]. However, traditional homology-based methods and deep learning models designed for other generic bioactive peptides may not generalize well to neuropeptide identification due to their high sequence heterogeneity, variable length, and complex post-translational modifications [8-10]. Therefore, developing reliable computational methods for efficient, large-scale discovery of neuropeptides is imperative.

A cornerstone of neuropeptide prediction has been the use of traditional machine learning. This approach centers on pairing classical classifiers with manually engineered features. For instance, an early predictor, NeuroPIpred, utilized a support vector machine based on four classes of physicochemical features [11]. Subsequent work has developed along two principal directions. One path aims to construct more sophisticated models through strategies such as stacking and ensemble learning, as illustrated by PredNeuroP's dual-layer framework [12]. The other path focuses on optimizing the input data via feature selection, a technique utilized by NeuroPpred-Fuse and NeuroPred-FRL with their single-step and two-step strategies, respectively [13, 14]. Despite their straightforward nature, these features can only produce shallow sequence representations, thereby failing to capture the deep contextual information required for accurate prediction.

The paradigm for neuropeptide prediction has recently shifted towards deep learning. Liu et al. spearheaded this transition with an early breakthrough, NeuroCNN_GNB. Their model, which integrated a convolutional neural network with Gaussian Naive Bayes, was significant for proving the viability of automatic feature extraction [15]. With the rise of pre-trained protein language models (PLMs), including ESM, BERT, and ProtTrans, further progress has been achieved in this field. These approaches extract informative latent features from protein and peptide sequences and have been successfully applied to diverse proteomics and peptide-related prediction tasks [16, 17]. Specifically, although NeuroPred-PLM [18] combined ESM embeddings with a multi-

scale convolutional network, its simplistic feature aggregation scheme limited its ability to capture deep sequential dependencies. Likewise, NeuroPred-SVM [19] employed a shallow classifier that struggled to fully exploit hierarchical interactions from BERT embeddings. To enrich representations, fusion strategies like NeuroPred-AIMP [20] and NeuroPpred-SHE [21] integrated heterogeneous descriptors; however, they remained limited by simple feature concatenation, which often introduced redundancy and noise without adaptively selecting optimal signals. Furthermore, even the recent state-of-the-art framework, NeuroPpred-MSN [22], which aligned diverse representations via a Siamese network and contrastive learning, still lacked an explicit gated filtering mechanism. Collectively, these architectural bottlenecks underscore the necessity for a framework that synergizes deep multi-scale context extraction with adaptive feature purification.

To overcome the reliance of existing methods on unimodal feature representations and to mitigate the issue of information redundancy in multi-feature fusion, we propose NeuroPpred-PSCG, a novel multimodal deep learning framework. The core of this framework is the sophisticated integration of two complementary feature types: (1) Global semantic information, which encodes relationships between distant amino acids and evolutionary context using a pre-trained protein language model (ProtT5); (2) Local conformational features, which represent the local spatial arrangement of the sequence via predicted secondary structure information. To ensure the effective fusion of this heterogeneous information, we design a two-stage integration module. First, a bidirectional cross-attention mechanism facilitates deep interaction and alignment between the two feature modalities. Subsequently, a gated fusion mechanism adaptively assigns weights to the features, which allows for the dynamic prioritization of information most crucial for prediction. On this basis, a multi-scale gated convolutional network is applied to extract contextual patterns from the fused features, thereby boosting their expressive capacity. Finally, an attention pooling layer aggregates residue-level features into a sequence-level representation for classification. NeuroPpred-PSCG achieves competitive performance on the independent test set. Furthermore, extensive ablation studies, robustness analyses, and interpretability analyses confirm the model's effective discriminative ability and its potential for practical applications.

2 MATERIALS AND METHODS

2.1 Datasets

The study employs the public benchmark dataset previously established by Wang et al. to ensure a fair and comparable assessment [18]. In their data construction pipeline, the positive samples consisted of 11,282 experimentally confirmed neuropeptides collected from the NeuroPep 2.0 database [23]. These sequences underwent a two-stage filtering process: first, sequences of length 5 to 100 amino acids were retained; subsequently, the CD-HIT tool (version 4.8.1) was utilized to eliminate redundancy at a 90% sequence identity cutoff [24], resulting in 4463 non-redundant neuropeptide sequences. For dataset balancing, negative samples were randomly extracted by Wang et al. from the UniProt database and then processed using the identical procedure to ensure matching length distributions and redundancy removal. In total, the dataset contains 8926 sequences and was divided into training and independent testing subsets at a 9:1 split, which we strictly adhere to in our evaluation to ensure an unbiased and standard comparison.

Additionally, to evaluate the model's generalization under class-imbalanced conditions, we employ two imbalanced datasets from Wen et al. [21]. These datasets originate from the same source as the benchmark data but were constructed by the original authors using a more stringent 0.3 similarity cutoff for deduplication, yielding 454 positive samples and 2,542 negative samples (a ratio of approximately 1:5.6). This collection was then partitioned at a 9:1 ratio to form a training dataset and an imbalanced testing dataset. The external independent test set was specifically compiled from neuropeptides published after 2023 in the NCBI and PDB databases. Processed via the same pipeline, it contains 115 positive and 644 negative samples. Models are trained on the imbalanced training set and evaluated on both independent test sets. All datasets are summarized in Table 1.

2.2 Overview of the Model

As illustrated in Figure 1, NeuroPpred-PSCG employs a multimodal deep learning framework to achieve accurate neuropeptide identification. The model's input layer processes two heterogeneous feature modalities in parallel. The first modality utilizes the pre-trained protein language model ProtT5 to capture global semantic representations and latent evolutionary patterns from the sequence [25]. The second

integrates local conformational features derived from the secondary structure, as predicted by the PHAT web server [26]. Subsequently, these two modalities undergo deep interaction and alignment through a bidirectional cross-attention mechanism. This is followed by a linear gated fusion unit that performs adaptive weighting to emphasize the most discriminative information. Based on the fused features, a multi-scale gated convolutional network is introduced to model hierarchical contextual dependencies, thereby enhancing their expressive capacity. Finally, an attention pooling layer aggregates sequence-level features for the final prediction.

2.2.1 ProtT5 Semantic Embedding

To capture context-aware features from neuropeptide sequences, we utilize the pre-trained protein language model ProtT5-XL-UniRef50 (ProtT5) [25]. The model contains 33 Transformer layers and is pre-trained on 650 million UniRef50 protein sequences via a masked language modeling objective. Through this pre-training, the model can implicitly learn evolutionary information and the relationships between sequence, structure, and function. In practice, for any given amino acid sequence S, ProtT5 generates a 1024-dimensional embedding for each residue. To align these raw embeddings for the cross-modal interaction module, we employ a linear layer to map them into a shared 320-dimensional latent space, based on the hyperparameter optimization described below. This procedure significantly reduces computational costs while accentuating the semantic information pertinent to our prediction task.

2.2.2 Secondary Structure Embedding

For this work, we employ the PHAT web server to predict the secondary structure of neuropeptide sequences [26]. The PHAT model integrates a powerful pre-trained protein language model with a hypergraph-based multi-head attention architecture, which effectively transfers semantic information from large proteins to peptide sequences. It can generate predictions in both three-state (H: alpha-helix, E: extended beta-strand, C: coil/others) and eight-state (H: alpha-helix, G: 310-helix, I: π-helix, E: extended beta-strand, B: isolated beta-strand, T: turn, S: bend, C: coil/others) formats [27, 28]. For each generated secondary structure sequence, we first convert it into an initial structural vector using token and positional embedding techniques. Subsequently, a linear projection layer maps this structural vector to a dimensional space consistent with that of the reduced-dimension ProtT5 embeddings. This step performs feature-dimension alignment before cross-modal fusion and yields the final secondary structure representation used by the gated cross-attention module.

2.2.3 Gated Cross-Attention Mechanism

NeuroPpred-PSCG introduces a gated cross-attention mechanism (Figure 1) to foster deep integration between ProtT5 semantic embeddings and secondary structure features [29]. Unlike simple feature addition or concatenation, this mechanism leverages both bidirectional cross-attention and a linear gate to enable reciprocal enhancement. Specifically, deep semantic information from ProtT5 embeddings strengthens the secondary structure characterization, while local conformational details from the secondary structure, in turn, provide the embeddings with rich structural information. The features are then passed through the linear gating unit, which dynamically adjusts and combines the information flow from the two modalities for efficient and adaptable fusion.

(1) Cross-Attention Mechanism

Cross-attention extends the standard attention mechanism, specifically designed to model the interdependencies between two heterogeneous feature sequences [30]. It enables one feature sequence, acting as the Query, to dynamically aggregate relevant information from another sequence, which serves as the Key and Value, thereby facilitating cross-modal alignment. Specifically, the bidirectional interaction is driven by two explicit information flows. In the ProtT5-to-ss8 flow, ss8 features act as the Query, and ProtT5 embeddings serve as the Key and Value, enabling structural features to integrate global semantic context. Conversely, in the ss8-to-ProtT5 flow, ProtT5 embeddings query ss8 features to incorporate local structural cues. These dual pathways facilitate effective alignment between semantic and structural representations. The cross-attention operation follows Equations (1) and (2).

$$\begin{aligned} head_i &= CrossAttention\left(Q_i, K_j, V_j\right) \\ &= Softmax\left(\frac{Q_i K_j^T}{\sqrt{d_k}}\right) V_j \end{aligned} \tag{1}$$

$$MHA = concat\left(\left[head_1, head_1, \mathrm{K}, head_h\right]\right) W_O \tag{2}$$

where the Q, K, and V matrices are generated from the ProtT5 embeddings and secondary structure features, respectively. The term W_O is a learnable weight matrix. The term d_K corresponds to the key dimension, whereas h denotes the number of attention heads.

(2) Linear Gating Fusion Mechanism

A linear gating unit is employed to fuse the bidirectional cross-attention-enhanced

features h_{ProtT5}, h_{ss8}. This unit consists of a linear transformation and a Softmax function, which dynamically compute a set of complementary gates. These gates then modulate the input features in a weighted sum to yield the final fused representation h_{fuse}, as defined in Equations (3) and (4).

$$[w_1, w_2] = Softmax\left(h_{ProtT5}W + h_{ss8}W\right) \tag{3}$$

$$h_{fuse} = w_1 \cdot h_{ProtT5} + w_2 \cdot h_{ss8} \tag{4}$$

where w_1 and w_2 represent the gating weights for the ProtT5 features and the secondary structure features, respectively.

2.2.4 Multi-Scale Gated Convolution Network

To further model hierarchical contextual relationships in the fused features, we devise a multi-scale gated convolutional module [31]. This module comprises a set of parallel gated convolutional units, with every unit utilizing a distinct kernel size to cover a distinct receptive field. Specifically, an input feature *X* is processed by each unit through two independent convolutional paths to generate the candidate features *C* and the gating signal *G*, respectively. Equation (5) gives the candidate feature and gating signal for each gated convolutional unit.

$$\begin{aligned} C &= Conv1D_c\left(X\right) = X \cdot W_c + b_c \\ G &= Conv1D_g\left(X\right) = X \cdot W_g + b_g \end{aligned} \tag{5}$$

where W_c, W_g and b_c, b_g denote the weight and bias matrices associated with the one-dimensional convolution, respectively.

Subsequently, the gating signal *G* is passed through a Sigmoid activation function, which scales its values to the range [0, 1], thereby generating gating weights that are applied element-wise to the candidate features *C*. This mechanism enables the model to regulate information flow by selecting context-relevant features while attenuating less informative ones. The result is then sequentially processed by a ReLU activation function followed by a dropout layer, to introduce nonlinearity and enhance the model's generalization. The gated convolutional output is expressed in Equation (6).

$$Z = Dropout\left(ReLU\left(C \text{ e } Sigmoid\left(G\right)\right)\right) \tag{6}$$

2.2.5 Output layer

The feature matrix *Z*, produced by the gated convolutional module, captures both local and long-distance sequential relationships. We employ an attention pooling strategy to compress this high-dimensional feature matrix into a sequence-level vector *z* [32]. This approach not only lowers computational complexity but also enables the

model to assign different importance to positions within the sequence. Consequently, features that are more relevant to classification receive greater emphasis. Equations (7) and (8) describe the calculation of attention weights and the sequence-level representation, respectively.

$$\alpha_i = \frac{\exp\left(w^T \tanh\left(z_i\right)\right)}{\sum_{j=1}^{L} \exp\left(w^T \tanh\left(z_j\right)\right)}, i = 1,\mathrm{K}\ ,L \tag{7}$$

$$z = \sum_{i=1}^{L} \alpha_i z_i \tag{8}$$

where $Z = \left[z_1, z_2, \mathrm{K}\ , z_L\right]^T \in R^{L \times d}$, L represents the sequence length.

Finally, the vector z is passed through a two-layer fully connected classification head to obtain the prediction. The classification output is obtained using Equation (9).

$$\hat{y} = Softmax\left(W_2 \cdot ReLu\left(W_1 \cdot z + b_1\right) + b_2\right) \tag{9}$$

2.3 Fast Gradient Method (FGM) for Robustness Evaluation

During neuropeptide sequencing, experimental noise and sequencing errors may perturb the input sequence, potentially degrading the model's stability and generalizability. To investigate this issue, we adopt the fast gradient method (FGM) [33, 34]. This approach generates adversarial samples by adding small perturbations along the gradient direction of the loss function, which serve to examine the model's sensitivity to minor input alterations.

In practice, we apply FGM directly to the embedding space by perturbing the projection layer of ProtT5-derived features. Given an input x with label y, and the loss function $J\left(x, y\right)$, the perturbation is computed by Equations (10) and (11).

$$\delta = \varepsilon \cdot sign\left(\nabla_x J\left(\theta, x, y\right)\right) \tag{10}$$

$$x_{attack} = x + \delta \tag{11}$$

where ε represents a hyperparameter governing the perturbation's size, and $sign\left(\cdot\right)$ refers to the sign function, which is used to extract the gradient direction in order to generate adversarial examples with maximal perturbation effect.

2.4 Evaluation Criteria

Following previous studies, we employ eight performance indicators: accuracy (ACC), specificity (SP), sensitivity (SN), precision (Pre), F1-score (F1), Matthews correlation coefficient (MCC), area under the ROC curve (AUROC), and area under

the PR curve (AUPRC). Among these, SN and Pre quantify the completeness and correctness of positive predictions, respectively, while their harmonic mean, the F1-score, serves as a unified indicator for the positive class. In contrast, SP reflects performance on the negative samples. MCC is reported for its stability under class imbalance, and AUROC and AUPRC together provide threshold-free measures of the model's overall classification capability. These metrics are defined in Equation (12).

$$
\begin{aligned}
ACC &= \frac{TP + TN}{TP + TN + FP + FN} \\
SP &= \frac{TN}{TN + FP} \\
SN &= \frac{TP}{TP + FN} \\
Pre &= \frac{TP}{TP + FP} \\
F1 &= 2\frac{Pre \times SN}{Pre + SN} \\
MCC &= \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FN)(TN + FP)}}
\end{aligned}
\tag{12}
$$

3 RESULTS AND DISCUSSION

3.1 Performance Comparison with Existing Models

This study compares NeuroPpred-PSCG with eight representative neuropeptide prediction methods using the unified independent test set. For consistency and transparency, the baseline results are taken directly from the original publications, which used the same benchmark datasets and train/test partitions [18, 21]. This avoids potential bias from unofficial reimplementations or suboptimal hyperparameter tuning. As shown in Table 2, NeuroPpred-PSCG outperforms competing methods across multiple metrics. Specifically, it achieves the highest scores in ACC, SN, F1-score, and MCC, reaching 94.8%, 95.1%, 94.8%, and 0.896, respectively. Relative to the current state-of-the-art model NeuroPpred-MSN, NeuroPpred-PSCG improves ACC and MCC by 1.2% and 0.024, respectively, reflecting stronger discriminative capability. It is worth noting that NeuroPred-FRL reports the highest precision (96.0%), but its sensitivity remains limited (75.7%), implying a higher rate of missed positives. In contrast, NeuroPpred-PSCG achieves the highest sensitivity (95.1%) while maintaining a competitive precision (94.6%), resulting in a more balanced trade-off between detecting positive samples and controlling false positives. Furthermore, NeuroPpred-

PSCG attains an AUROC of 0.980, which is comparable to the best value reported by NeuroPred-MSN (0.983). This suggests that the model maintains consistent performance across varying classification thresholds. To validate the performance differences, we apply the Friedman test and pairwise Wilcoxon signed-rank tests. The Friedman test reveals significant overall differences among the models ($p < 0.001$). Subsequent pairwise analysis demonstrates that NeuroPpred-PSCG significantly outperforms NeuroPred-FRL ($p = 0.0313$) and shows a marginally significant improvement over NeuroPpred-MSN at the 0.10 level ($p = 0.0781$). For the remaining six methods, our model demonstrates significant superiority (all $p = 0.0156$). Together, these results provide statistical support for the superior performance of our model.

3.2 Exploration of the NeuroPpred-PSCG Architecture

To refine the architecture of NeuroPpred-PSCG, we perform a systematic hyperparameter analysis covering both the choice of pre-trained backbone and key architectural settings. The initial investigation focuses on four prominent protein language models: ESM-2, ProtT5, ProtBERT, and ProtALBERT [25, 35]错误!未找到引用源。. As illustrated in Figure 2, ProtT5 consistently achieves the best performance across all metrics, indicating its strong capability in capturing informative sequence representations. ESM-2 ranks second, with ACC of 93.9% and MCC of 0.878. In contrast, ProtBERT and ProtALBERT show inferior performance, with ProtBERT exhibiting the most noticeable decline (a reduction of 3.5% in ACC and 0.069 in MCC relative to ProtT5). These results underscore the importance of ProtT5 for downstream prediction tasks.

Using ProtT5 as the backbone, we further tune key architectural hyperparameters. Table 3 shows that the projection dimension of 320 achieves the strongest overall performance (ACC: 94.8%, F1-score: 94.8%, MCC: 0.896). Increasing the dimension to 384 slightly improves sensitivity (95.7%) but negatively affects overall performance, while a larger dimension (512) leads to an additional drop. This suggests that moderate dimensionality better balances representation capacity and redundancy. For the multi-scale gated CNN, the kernel configuration (3, 5) yields optimal results, while larger combinations, including (5, 7) and (7, 9), reduce overall performance. We also examine the number of attention heads h in the gated cross-attention module. Among $h = 2, 4, 8$, and 16, $h = 8$ achieves the best overall performance. Reducing h to 2 decreases ACC and MCC to 93.6% and 0.872, respectively, whereas increasing h to 16 reduces these values to 94.0% and 0.881. Overall, backbone choice and architectural tuning jointly dictate performance.

3.3 Ablation Experiments

A series of ablation studies is performed to systematically assess the contributions of the key modules within NeuroPpred-PSCG. We benchmark the performance of multiple model variants against the complete model (our baseline) on the independent test set, which allows us to verify the efficacy of the multimodal features and the cross-modal fusion mechanism.

3.3.1 Analysis of the Effectiveness of Multimodal Features

To investigate the contributions of each feature modality, we design three model variants: (i) one using only 3-state secondary structure information (designated as w/o ProtT5, w/ss3); (ii) one using only 8-state secondary structure information (w/o ProtT5, w/ss8); and (iii) one using only ProtT5 semantic features (w/o ss8, w/ProtT5). As illustrated in Figure 3A and Figure 3C, the model's performance degrades substantially upon the removal of ProtT5 features when relying solely on either ss3 or ss8 information. Specifically, the ACC drops to 72.3% and 72.6%, the MCC to 0.449 and 0.453, and the AUROC to 0.756 and 0.791, respectively. This indicates that structural information alone is insufficient to adequately capture the high-level semantic context of neuropeptide sequences. Conversely, when retaining only the ProtT5 semantic features, the model's performance recovers markedly, achieving an ACC of 94.3% and an MCC of 0.885. This finding highlights the importance of the pre-trained protein language model in capturing informative sequence patterns. Finally, the full model demonstrates superior performance for all indicators, providing strong evidence for the significant complementarity and synergy between semantic and structural information.

To further evaluate the cost-effectiveness of introducing ss8-derived structural information, we analyze model size and computational cost. Model size is characterized by trainable parameters and FP32 memory footprint, while computational cost is measured by MACs and FLOPs per sample. As shown in

Table 4, compared to the ProtT5-only variant (w/o ss8), the full NeuroPpred-PSCG model introduces a moderate increase in parameters (from 1.189 M to 1.827 M) and computational cost (from 0.249 G to 0.388 G FLOPs/sample). Importantly, this bounded overhead yields consistent performance gains, improving ACC from 94.3% to 94.8% and MCC from 0.885 to 0.896. As detailed in Table 5, the total overhead of integrating the ss8 modality includes 0.639 M parameters, 2.555 MB of FP32 storage, and 0.140 G FLOPs/sample. This additional cost stems primarily from the cross-

attention and gated fusion modules. In contrast, the ss8 embedding itself is remarkably lightweight, requiring only 0.022 M parameters and 0.004 G FLOPs/sample. Although ProtT5 remains the primary predictive source, the performance gains from structural cues justify this marginal complexity, confirming the efficiency of our multimodal architecture.

3.3.2 Contribution Analysis of the Cross-Modal Fusion Mechanism

To investigate the necessity of the cross-modal fusion mechanism, we construct the following variants: (i) a model with both the cross-attention and the gated convolutional network removed (designated as w/o crossatt, w/o gatecnn); (ii) one retaining only the cross-attention module (w/o gatecnn, w/ crossatt); and (iii) one retaining only the gated convolutional network (w/o crossatt, w/ gatecnn). As shown in Figure 3B, completely removing the fusion module leads to a significant performance drop, with the ACC decreasing from 94.8% to 93.2% and the MCC from 0.896 to 0.865. Further comparison reveals that the model with only cross-attention (ACC: 94.4%, MCC: 0.888) slightly outperforms the one with only the gated convolutional (ACC: 94.3%, MCC: 0.885), although both are inferior to the full NeuroPpred-PSCG model. This highlights the necessity of the gated convolutional, whose adaptive filtering mechanism effectively suppresses noise and refines the fused features, complementing the cross-modal interactions established by the attention module. The coordinated interaction of these two modules is essential for improving feature fusion, which in turn supports the underlying design of the NeuroPpred-PSCG architecture.

3.4 Evaluation of Model Robustness and Generalization

3.4.1 Robustness Analysis Under Embedding and Sequence-Level Perturbations

We analyze the robustness of NeuroPpred-PSCG using three rigorous tests: 10-fold cross-validation, FGM-based adversarial testing, and BLOSUM62-guided conservative substitutions. The cross-validation results (Table 6) indicate stable performance across different data splits, with ACC ranging from 93.4% to 94.9%. The average scores are 93.89% ± 0.44 for ACC, 93.90% ± 0.43 for F1-score, and 0.879 ± 0.009 for MCC. A high AUROC of 0.971 ± 0.015, together with small standard deviations, further indicates that the model is not sensitive to data partitioning. Under FGM perturbations (Table 7), the model performance remains largely unaffected. Only minor reductions are observed in ACC (94.8% to 94.5%), MCC (0.896 to 0.890), and AUROC (0.980 to 0.973), confirming the model's robustness against adversarial noise. This indicates that NeuroPpred-PSCG remains stable under data variability and input perturbations.

Subsequently, we evaluate sequence-level robustness on the independent test set using BLOSUM62-guided conservative amino acid substitutions. Specifically, 1%, 5%, and 10% of residues are randomly replaced with biochemically similar amino acids (BLOSUM62 scores $\geqslant$ 0), and each setting is repeated five times using fixed random seeds. For each mutated sequence set, ProtT5 and ss8 features are regenerated before prediction. Conservative substitutions are used because they are more likely to preserve the biochemical properties of short neuropeptides than insertions or deletions, which may alter peptide length, sequence motifs, and biological activity. As shown in

Table 8, NeuroPpred-PSCG maintains stable performance under 1%, 5%, and 10% conservative substitutions, with ACC values of 92.06%, 92.34%, and 91.22%, respectively, AUROC values above 0.967, and prediction consistency above 94.28%. Table 9 further shows that mutation sensitivity analysis yields small mean absolute probability shifts (0.041–0.053) and limited mean confidence drops (<0.046), suggesting that NeuroPpred-PSCG is reasonably stable under mild, biochemically constrained sequence-level variations.

3.4.2 Analysis of the Model's Generalization Ability on the Imbalanced Test Sets

To assess the generalization capability of NeuroPpred-PSCG on imbalanced datasets, we train and test our model on the dataset constructed by Wen et al. [21]. As detailed in Table 10 and Table 11, baseline metrics are directly adopted from their original publications. On the imbalanced independent test set, NeuroPpred-PSCG achieves strong overall performance in terms of ACC (96.0%), F1-score (87.0%), and MCC (0.846). Although PredNeuroP achieved the highest ACC (99.0%), its much lower SN (64.4%) indicates a bias toward the majority class. However, NeuroPpred-PSCG strikes a superior balance between SN (88.9%) and SP (97.2%). Although NeuroPpred-MSN reported a slightly higher AUROC (0.974), its lower MCC and F1-score suggest that its overall discriminative power and stability are inferior to our model. The Friedman test further confirms a significant overall difference among models ($p = 0.0015$). At a significance level of 0.1, Wilcoxon tests indicate that NeuroPpred-PSCG significantly outperforms NeuroPpred-SHE, NeuroPpred-Fuse, NeuroPred-FRL, and PredNeuroP, with p-values of 0.0547, 0.0234, 0.0156, and 0.0391, respectively. The comparisons with NeuroPpred-MSN and NeuroPred-PLM are not statistically significant (both $p = 0.1484$); however, the metric-level results indicate that NeuroPpred-PSCG achieves a better trade-off between positive-sample detection and negative-sample discrimination.

On the external imbalanced test set, NeuroPpred-PSCG also achieves the highest ACC (88.8%) and MCC (0.493), showcasing its robust, comprehensive discriminative ability. Although its F1-score is marginally lower than that of NeuroPpred-SHE (55.0%), NeuroPpred-PSCG maintains a lead in precision (73.4%) and specificity (97.4%), indicating a significant advantage in controlling false positives. Taken together, these results demonstrate that NeuroPpred-PSCG achieves a superior performance trade-off and greater robustness in highly imbalanced data scenarios. Similarly, the Friedman test reveals a significant overall performance difference among the models ($p < 0.001$). The post-hoc Wilcoxon tests show that NeuroPpred-PSCG significantly outperforms NeuroPred-PLM, NeuroPpred-Fuse, NeuroPred-FRL, and PredNeuroP, with all comparisons yielding $p = 0.0078$. The comparisons with NeuroPpred-MSN and NeuroPpred-SHE are not statistically significant, with p-values of 0.4688 and 0.3281, respectively. Nevertheless, NeuroPpred-PSCG achieves the best ACC, SP, Pre, and MCC, supporting its advantage in overall discrimination and false-positive control.

3.5 Visualization of Feature Learning Using UMAP

We employ uniform manifold approximation and projection (UMAP) 错误!未找到引用源。 to visualize feature representations from different stages of NeuroPpred-PSCG on the independent test set, aiming to examine their discriminative characteristics. This visualization highlights both the distribution of samples and the degree of class separation, while maintaining the intrinsic topology of the data. The results reveal that the initial secondary structure features (Figure 4A) possess limited discriminative power, with positive and negative samples largely mixed in the embedding space. Introducing ProtT5 semantic features (Figure 4B) improves class separation to some extent, though considerable overlap still exists. Further integration of these two modalities via bidirectional cross-attention (Figure 4C) leads to a more organized feature distribution, with clearer separation boundaries. Subsequently, the multi-scale gated convolutional network (Figure 4D) further refines these features, sharpening the inter-class boundary and promoting the formation of more compact and distinct clusters. Finally, after passing through the global attention pooling layer (Figure 4E), NeuroPpred-PSCG acquires a highly separable feature representation, where the two classes exhibit a clear decision boundary and excellent intra-class cohesion. These visualizations show that the model representations evolve from a weak, unimodal state to a structured, highly discriminative multimodal form, intuitively validating our framework's effectiveness and design.

3.6 Interpretability Analysis

3.6.1 Feature Importance Analysis

To interpret the model's predictions, we evaluate the contributions of ProtT5 and ss8 features using SHapley Additive exPlanations (SHAP) analysis 错误!未找到引用源。. As shown in Figure 5 and Figure 6, the top 10 most influential features all originate from ProtT5 embeddings. The dominance of ProtT5 is further evidenced by its mean absolute SHAP value (0.001412), which is significantly higher than that of the ss8 features (0.000199), demonstrating its leading role in driving model predictions at the global level. At the feature level, the SHAP distributions also reveal distinct directional effects. For instance, lower values (blue) in ProtT5_144, ProtT5_288, and ProtT5_84 drive positive predictions, whereas higher values (red) in ProtT5_277, ProtT5_54, and ProtT5_69 favor positive outcomes. Furthermore, the asymmetric and scattered distribution of these SHAP values implies that the model relies on non-linear, context-dependent interactions rather than simple monotonic trends. These results validate the strong discriminative capacity of ProtT5 embeddings for neuropeptide identification.

3.6.2 Attention-Based Interpretability Analysis

To investigate the internal decision-making mechanism of the NeuroPpred-PSCG model and evaluate its interpretability, we visualize its cross-attention weights and compare them with conserved motifs identified by MEME [38]. As illustrated in Figure 7, the attention weights are highly concentrated on the "FQGMR" region, and the high-attention region not only aligns precisely with the "WFGPR" motif identified by MEME but also significantly extends to cover the upstream "PGL" segment. This strong concordance between the key sequences autonomously learned by the model and known conserved motifs provides compelling evidence that NeuroPpred-PSCG not only exhibits excellent classification performance but can also accurately identify the critical sequence patterns that drive its decisions. The attention mechanism reveals the intrinsic rationale and biological basis of the model's predictions by assigning greater weights to the amino acid residues with higher contributions.

3.6.3 Mutation-Scan Analysis

To investigate residue-level interpretability, we perform in silico mutagenesis (ISM) on the independent test set. Each residue position is systematically substituted with the 20 standard amino acids, and the resulting mutant sequences are re-predicted. As shown in Figure 8A and Figure 8B, the sequence logos are generated using the kpLogo web server [39]. Positive samples are enriched with residues such as S, L, R, and G, whereas

negative samples show a clear enrichment of methionine (M) at the N-terminal position. The ISM heatmaps of three representative neuropeptides further demonstrate how these residue preferences influence model predictions (Figure 8C-E). Substituting the N-terminal residue with M markedly reduces the prediction score, consistent with its enrichment in non-neuropeptides. Similarly, substitutions to cysteine (C), particularly at internal positions, lead to low-score regions, agreeing with the low frequency of C in positive neuropeptide logos. To reduce sequence-specific bias, all positive samples are normalized into 10 position-ratio intervals, and a global ISM visualization is generated (Figure 8F). This analysis confirms that M exerts its strongest effect at the first position-ratio interval, while C shows strong effects across multiple internal-to-C-terminal intervals. Together, these results indicate that NeuroPpred-PSCG captures biologically meaningful positional residue preferences rather than relying on simple sequence memorization.

3.7 Limitations and Future Directions

Despite NeuroPpred-PSCG's competitive performance, neuropeptide identification remains more challenging than predicting other short bioactive peptides. Although prediction models developed for antiviral peptides (e.g., TargetAVP-DeepCaps [40]), antimicrobial peptides (e.g., deep-AMPpred [41]), and anticancer peptides (e.g., pACP-HybDeep [42]) often report higher evaluation metrics, these methods cannot be directly transferred to neuropeptide identification. Neuropeptides typically exhibit high sequence heterogeneity, weak evolutionary conservation, limited shared physicochemical motifs, distinct functional mechanisms, and complex post-translational processing. These factors complicate identification and suggest that cross-task performance comparisons should be interpreted cautiously. Given these challenges, NeuroPpred-PSCG's balanced performance confirms the effectiveness of integrating ProtT5-derived semantics with structural information via adaptive cross-modal fusion.

Nevertheless, several limitations remain and should be addressed in future work. Our future efforts will focus on the following four aspects: (1) Integration of 3D structural information: The current model is built upon sequence information and secondary structure predictions, without explicitly modeling the 3D conformations that are closely linked to biological function. Introducing structural data predicted by AlphaFold2 [43] or ESMFold [44] may help capture spatial arrangements and functional domains more effectively, which could contribute to improved prediction accuracy. (2) Generalization under external imbalanced conditions: While NeuroPpred-PSCG maintains competitive

overall performance on the external imbalanced test set, its sensitivity drops under external distribution shifts. This suggests that identifying positive neuropeptides remains challenging despite comparable class ratios, possibly due to variations in sequence composition and data sources. Future work will explore threshold calibration and cost-sensitive learning to enhance sensitivity and real-world robustness. (3) Parameter-efficient fine-tuning (PEFT): The present study relies on static ProtT5 embeddings, without further adapting the pre-trained model to the target task. Applying PEFT methods, such as LoRA [45], could enable efficient fine-tuning with a limited number of additional parameters. This could enhance its ability to capture task-relevant patterns in neuropeptide sequences, while preserving efficiency in computation and memory usage. (4) Generalization to a wider range of bioactive peptides: NeuroPpred-PSCG has demonstrated considerable versatility. We plan to extend to other bioactive peptide tasks, such as antimicrobial and anti-inflammatory peptide prediction, to explore its potential as a general-purpose framework.

4 CONCLUSIONS

This study introduces NeuroPpred-PSCG, a multimodal deep learning framework for neuropeptide identification. The model architecture integrates information across different scales. Through the application of a bidirectional cross-attention mechanism and a multi-scale gated convolutional network, it harmonizes the high-level semantic insights captured by the ProtT5 language model with the localized structural patterns defined by secondary structures, thus facilitating a hierarchical analysis of the sequence. Our method achieves consistent improvements over existing state-of-the-art methods in overall performance. Moreover, ablation results highlight the importance of the multimodal fusion strategy for model performance. Additional evaluations of robustness and generalization, together with visualization analyses, show that the model remains stable and captures biologically meaningful patterns. Overall, this study highlights the potential of multimodal deep fusion in characterizing complex biological sequences, offering a promising approach for the discovery of bioactive peptides.

Data and Software Availability

The dataset and source code for this study are publicly available at https://github.com/xinyizhang186/NeuroPpred-PSCG.

Author Contributions

Shengli Zhang conceived and supervised the study, and Xinyi Zhang conducted the computational analyses. Both authors drafted, revised, and approved the manuscript.

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Conflict of Interest Disclosure

The authors declare no conflicts of interest.

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Table Captions:

Table 1. Details of the datasets.

Table 2. Performance comparison with existing models on the independent test set.

Table 3. Performance comparison of different projection dimensions, kernel sizes, and attention heads.

Table 4. Complexity and performance comparison between the w/o ss8 variant and

NeuroPpred-PSCG.

Table 5. Additional complexity from the ss8 branch and cross-modal fusion module.

Table 6. 10-fold cross-validation performance of NeuroPpred-PSCG.

Table 7. Robustness evaluation of NeuroPpred-PSCG under FGM perturbation.

Table 8. Robustness performance under conservative substitution perturbations.

Table 9. Prediction probability stability under conservative substitution perturbations.

Table 10. Comparison with existing models on the imbalanced independent test set.

Table 11. Comparison with existing models on the external imbalanced test set.

Table 1. Details of the datasets.

Dataset	Train-Test	Positive	Negative
Benchmark dataset	Train	4019	4019
	Test	444	444
Imbalanced dataset	Train	409	2290
	Test	45	252
External test dataset	Test	115	644

Table 2. Performance comparison with existing models on the independent test set.

Model	ACC(%)	SN(%)	Pre(%)	F1(%)	MCC	AUROC
NeuroPpred-PSCG	**94.8**	**95.1**	94.6	**94.8**	**0.896**	0.980
NeuroPpred-MSN[22]	93.6	92.3	94.7	93.5	0.872	**0.983**
NeuroPred-AIMP[20]	92.3	92.6	92.2	92.4	0.847	0.974
NeuroPpred-SHE[21]	92.7	92.8	92.6	92.7	0.854	0.978
NeuroPred-PLM[18]	92.2	94.1	90.7	92.1	0.845	0.968
NeuroPpred-Fuse[13]	90.5	90.8	90.6	90.7	0.813	0.934
NeuroPred-FRL[14]	86.1	75.7	**96.0**	84.7	0.740	0.921
PredNeuroP[12]	86.4	78.2	93.5	85.2	0.738	0.931
NeuroPIpred[11]	73.2	75.5	72.2	73.8	0.464	0.810

Table 3. Performance comparison of different projection dimensions, kernel sizes, and attention heads.

Parameter settings		ACC(%)	SN(%)	Pre(%)	F1(%)	MCC	AUROC
Projection dim	256	94.1	93.9	94.3	94.1	0.883	**0.981**

	320	**94.8**	95.1	**94.6**	**94.8**	**0.896**	0.980
	384	94.4	**95.7**	93.2	94.4	0.888	0.961
	512	94.0	94.8	93.4	94.1	0.881	0.962
Kernel sizes	(3, 5)	**94.8**	95.1	**94.6**	**94.8**	**0.896**	**0.980**
	(5, 7)	94.1	93.7	**94.6**	94.1	0.883	0.964
	(7, 9)	94.3	**95.7**	93.0	94.3	0.886	0.968
Heads	2	93.6	92.6	94.5	93.5	0.872	0.977
	4	94.1	94.8	93.6	94.2	0.883	0.979
	8	**94.8**	**95.1**	**94.6**	**94.8**	**0.896**	**0.980**
	16	94.0	94.8	93.4	94.1	0.881	0.971

Table 4. Complexity and performance comparison between the w/o ss8 variant and NeuroPpred-PSCG.

Model	**Params(M)**	**FLOPs/sample(G)**	**ACC(%)**	**MCC**
w/o ss8	1.189	0.249	94.3	0.885
NeuroPpred-PSCG	1.827	0.388	94.8	0.896

Table 5. Additional complexity from the ss8 branch and cross-modal fusion module.

Added component	**Params (M)**	**FP32 size (MB)**	**MACs/ sample(M)**	**FLOPs/ sample(G)**
ss8 embedding+projection	0.022	0.089	2.048	0.004
Additional cross-attention	0.411	1.644	47.360	0.095
Cross-modal gated fusion	0.205	0.820	20.480	0.041
Fusion LayerNorm	0.001	0.003	0.000	0.000
Total added overhead	0.639	2.555	69.888	0.140

Table 6. 10-fold cross-validation performance of NeuroPpred-PSCG.

Folds	**ACC(%)**	**SN(%)**	**Pre(%)**	**F1(%)**	**MCC**	**AUROC**
1	93.8	93.8	93.8	93.8	0.876	0.980
2	94.0	95.0	93.2	94.1	0.881	0.950

3	94.9	94.5	95.2	94.9	0.898	0.983
4	94.3	92.8	95.6	94.2	0.886	0.981
5	94.0	95.0	93.2	94.1	0.881	0.979
6	93.5	93.5	93.5	93.5	0.871	0.977
7	93.7	91.5	95.6	93.5	0.874	0.986
8	93.4	95.8	91.5	93.6	0.869	0.944
9	93.4	92.5	94.2	93.4	0.868	0.950
10	94.2	93.0	95.2	94.1	0.883	0.978
Mean	93.89±0.44	93.82±1.27	93.98±1.26	93.90±0.43	0.879±0.009	0.971±0.015

Table 7. Robustness evaluation of NeuroPpred-PSCG under FGM perturbation.

Model	ACC(%)	SN(%)	Pre(%)	F1(%)	MCC	AUROC
NeuroPpred-PSCG	94.8	95.1	94.6	94.8	0.896	0.980
FGM Attacked	94.5	96.0	93.2	94.6	0.890	0.973

Table 8. Robustness performance under conservative substitution perturbations.

Perturbation	ACC (%)	MCC	AUROC	Consistency (%)
Original	94.8	0.896	0.980	-
CS-1%	92.06 ± 0.56	0.840 ± 0.011	0.971 ± 0.002	95.02 ± 0.69
CS-5%	92.34 ± 0.26	0.847 ± 0.005	0.972 ± 0.002	95.63 ± 0.54
CS-10%	91.22 ± 0.48	0.825 ± 0.010	0.967 ± 0.002	94.28 ± 0.92

*CS denotes BLOSUM62-guided conservative amino acid substitution.

Table 9. Prediction probability stability under conservative substitution perturbations.

Perturbation	Mean |ΔP|	P95 |ΔP|	Mean confidence drop	|ΔP|>0.1(%)
CS-1%	0.043 ± 0.004	0.336 ± 0.039	0.037 ± 0.004	8.65 ± 0.57
CS-5%	0.041 ± 0.002	0.295 ± 0.022	0.034 ± 0.003	8.58 ± 0.35
CS-10%	0.053 ± 0.005	0.436 ± 0.100	0.046 ± 0.006	10.41±0.41

*CS denotes BLOSUM62-guided conservative amino acid substitution.

Table 10. Comparison with existing models on the imbalanced independent test set.

Model	ACC(%)	SP(%)	SN(%)	Pre(%)	F1(%)	MCC	AUROC
NeuroPpred-PSCG	96.0	**97.2**	**88.9**	85.1	**87.0**	**0.846**	0.917
NeuroPpred-MSN[22]	94.6	96.0	86.7	79.6	83.0	0.799	**0.974**
NeuroPpred-SHE[21]	93.9	96.8	77.7	81.4	79.5	0.760	0.962
NeuroPred-PLM[18]	93.6	96.8	75.5	**89.0**	78.1	0.744	0.950
NeuroPpred-Fuse[13]	91.9	94.4	77.7	71.4	74.4	0.697	0.951
NeuroPred-FRL[14]	92.9	96.4	73.3	78.5	75.8	0.717	0.919
PredNeuroP[12]	**99.0**	95.6	64.4	72.5	68.2	0.631	0.893

Table 11. Comparison with existing models on the external imbalanced test set.

Model	ACC(%)	SP(%)	SN(%)	Pre(%)	F1(%)	MCC	AUROC
NeuroPpred-PSCG	**88.8**	**97.4**	40.9	**73.4**	52.5	**0.493**	0.740
NeuroPpred-MSN[22]	88.5	97.0	**49.0**	71.2	51.9	0.483	**0.872**
NeuroPpred-SHE[21]	88.1	96.7	40.0	68.6	**55.0**	0.464	0.857
NeuroPred-PLM[18]	78.7	93.0	13.9	25.0	16.5	0.050	0.569
NeuroPpred-Fuse[13]	79.3	89.9	20.0	26.1	22.6	0.110	0.633
NeuroPred-FRL[14]	88.0	91.7	20.0	32.0	24.0	0.140	0.612
PredNeuroP[12]	82.0	93.1	20.0	34.3	25.2	0.166	0.652

Figure Captions:

Figure 1. Flowchart of the NeuroPpred-PSCG model. (A) Dataset construction and preprocessing. (B) Feature representation based on ProtT5 embeddings and predicted secondary-structure features. (C) Gated cross-attention mechanism enabling bidirectional cross-modal interaction and adaptive fusion. (D) Multi-scale gated convolution module for extracting hierarchical contextual features. (E) Classification module for neuropeptide identification.

Figure 2. Performance comparison of different pre-trained protein language models.

Figure 3. Results of the ablation study. (A) Effectiveness of multimodal features. (B)

Contribution of cross-modal fusion mechanisms. (C) ROC and PR curves.

Figure 4. Feature visualizations on the independent test set. (A) SS8 features. (B) ProtT5 embeddings. (C) Gated cross-attention features. (D) Multi-scale gated convolution features. (E) Global attention pooling features.

Figure 5. SHAP analysis for model interpretability of the top 10 features.

Figure 6. SHAP-based contribution analysis of ProtT5 and ss8 features.

Figure 7. Interpretability of the model's attention mechanism. The attention heatmaps demonstrate strong alignment with the top two sequence motifs identified by the MEME analysis. (A) "FQGMR". (B) "WFGPR".

Figure 8. Mutation-scan analysis. (A, B) Position-specific amino acid preferences for neuropeptides and non-neuropeptides visualized by kpLogo. (C-E) ISM heatmaps of three representative peptides, with colors indicating normalized prediction scores from low (red) to high (blue). (F) Global ISM visualization illustrating amino acid contribution patterns across position-ratio intervals.

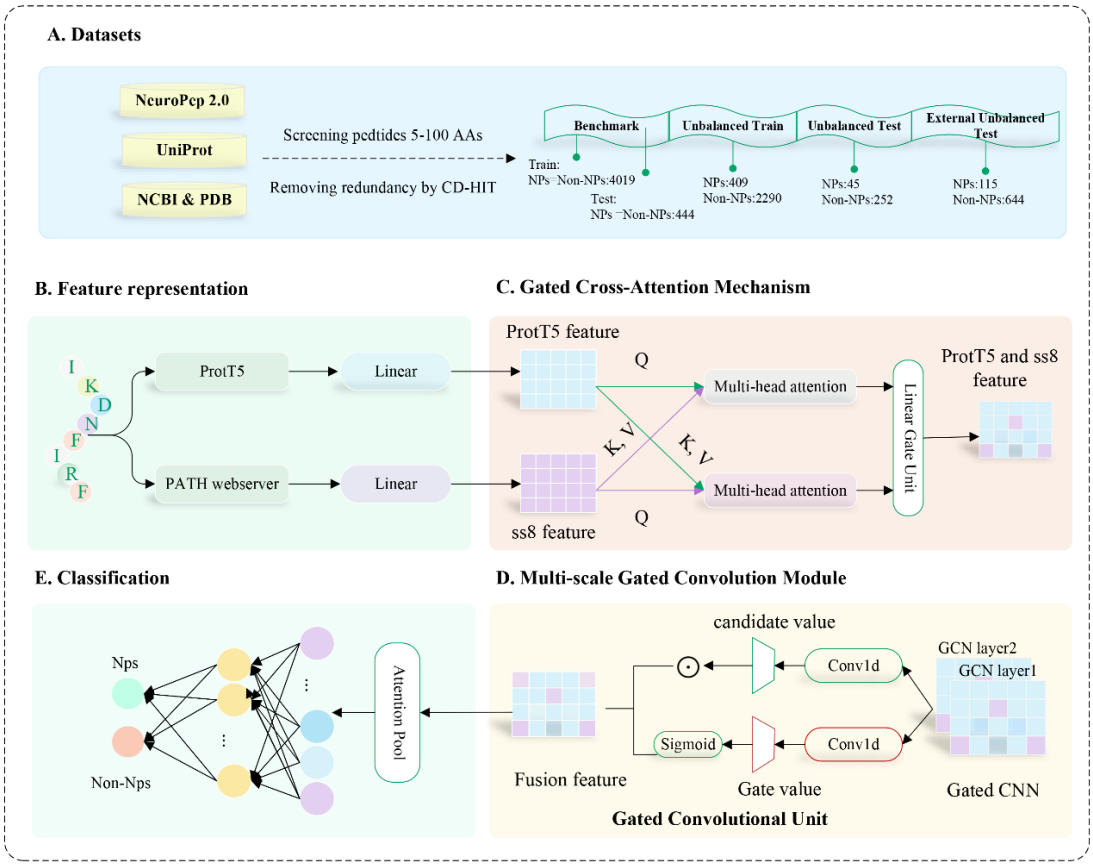


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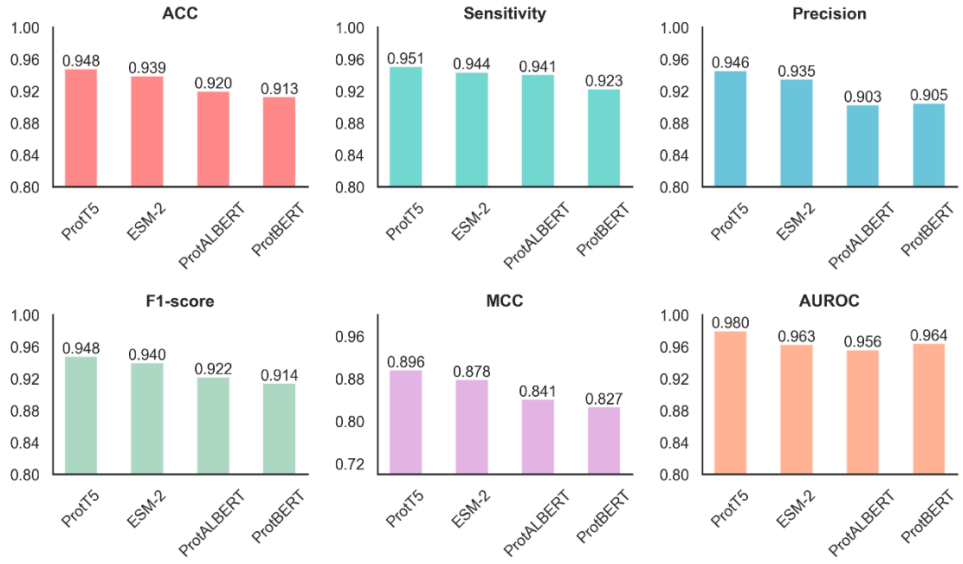


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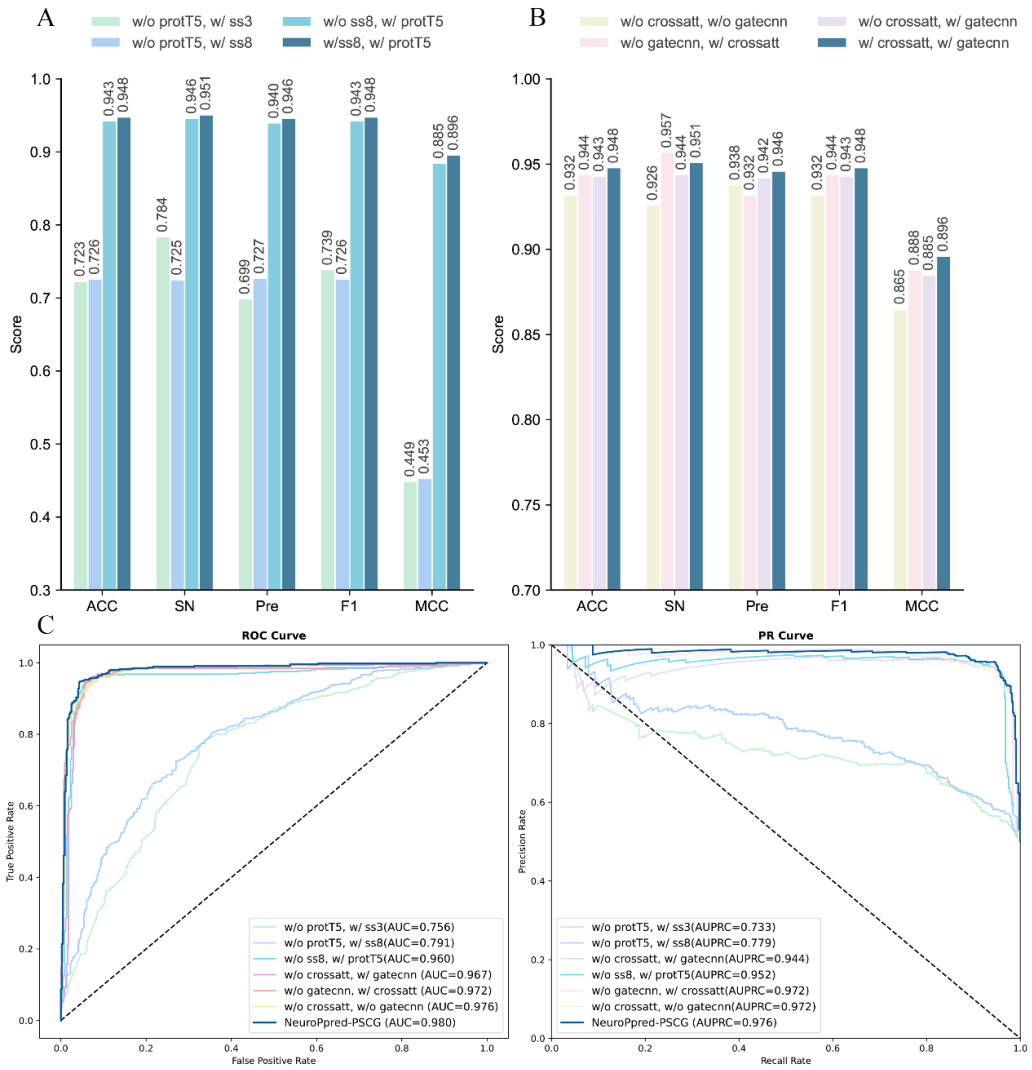


Figure 3. Results of the ablation study. (A) Effectiveness of multimodal features. (B) Contribution of cross-modal fusion mechanisms. (C) ROC and PR curves.

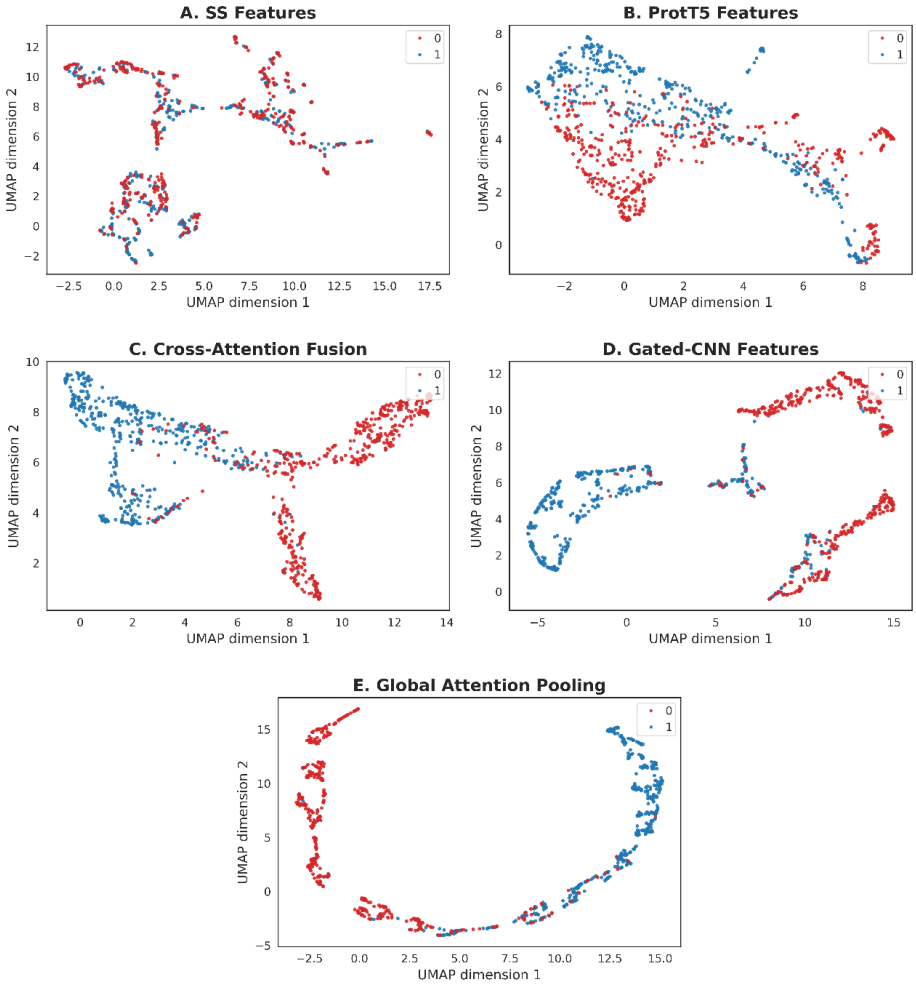


Figure 4. Feature visualizations on the independent test set. (A) SS8 features. (B) ProtT5 embeddings. (C) Gated cross-attention features. (D) Multi-scale gated convolution features. (E) Global attention pooling features.

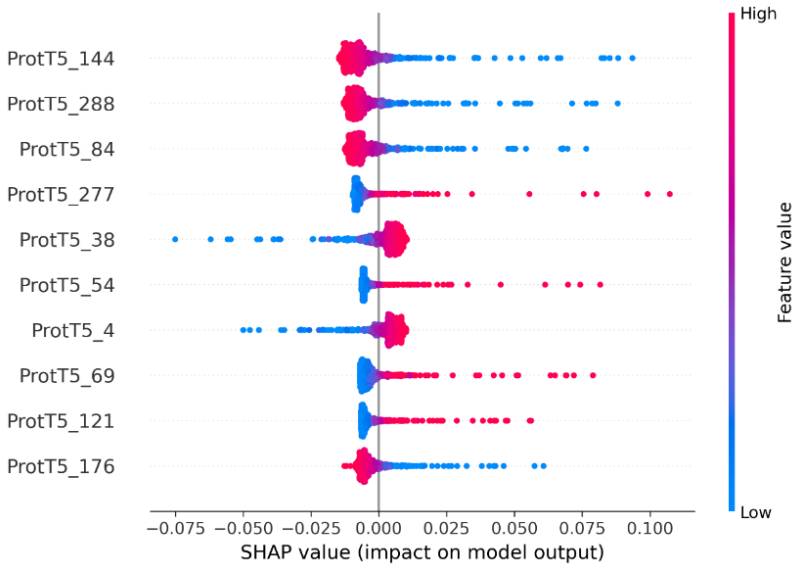


Figure 5. SHAP analysis for model interpretability of the top 10 features.

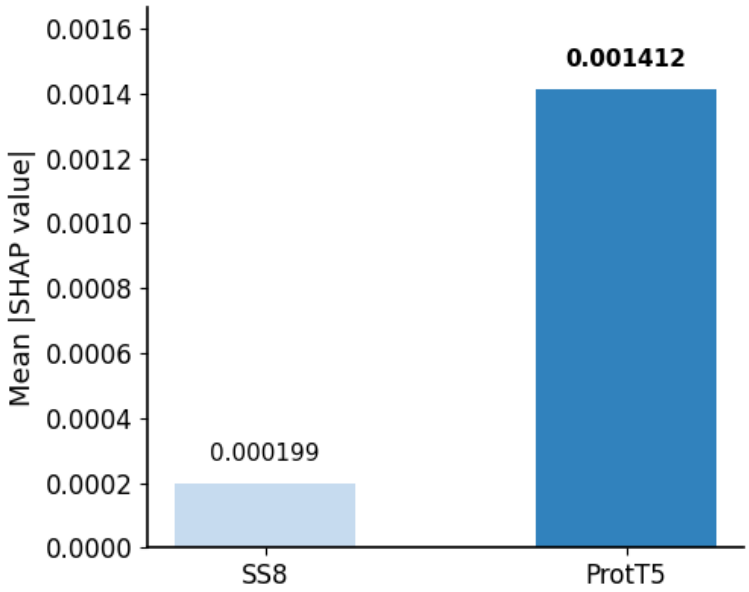


Figure 6. SHAP-based contribution analysis of ProtT5 and ss8 features.

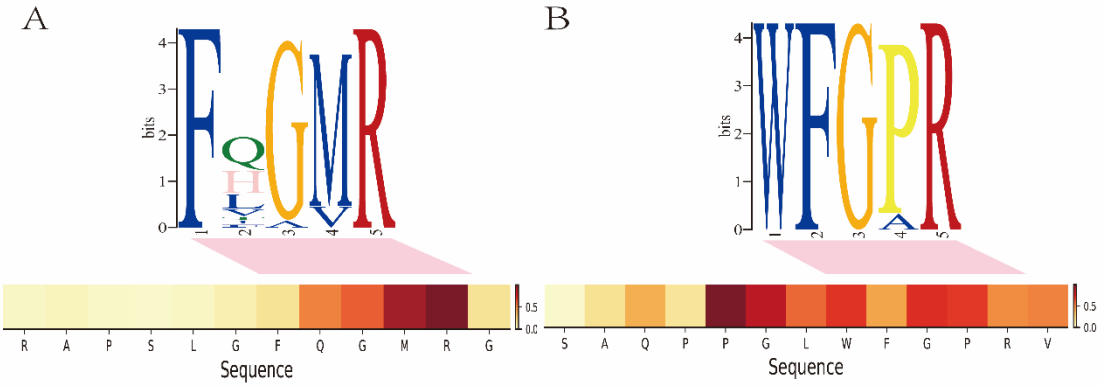


Figure 7. Interpretability of the model's attention mechanism. The attention heatmaps demonstrate strong alignment with the top two sequence motifs identified by the MEME analysis. (A) "FQGMR". (B) "WFGPR".

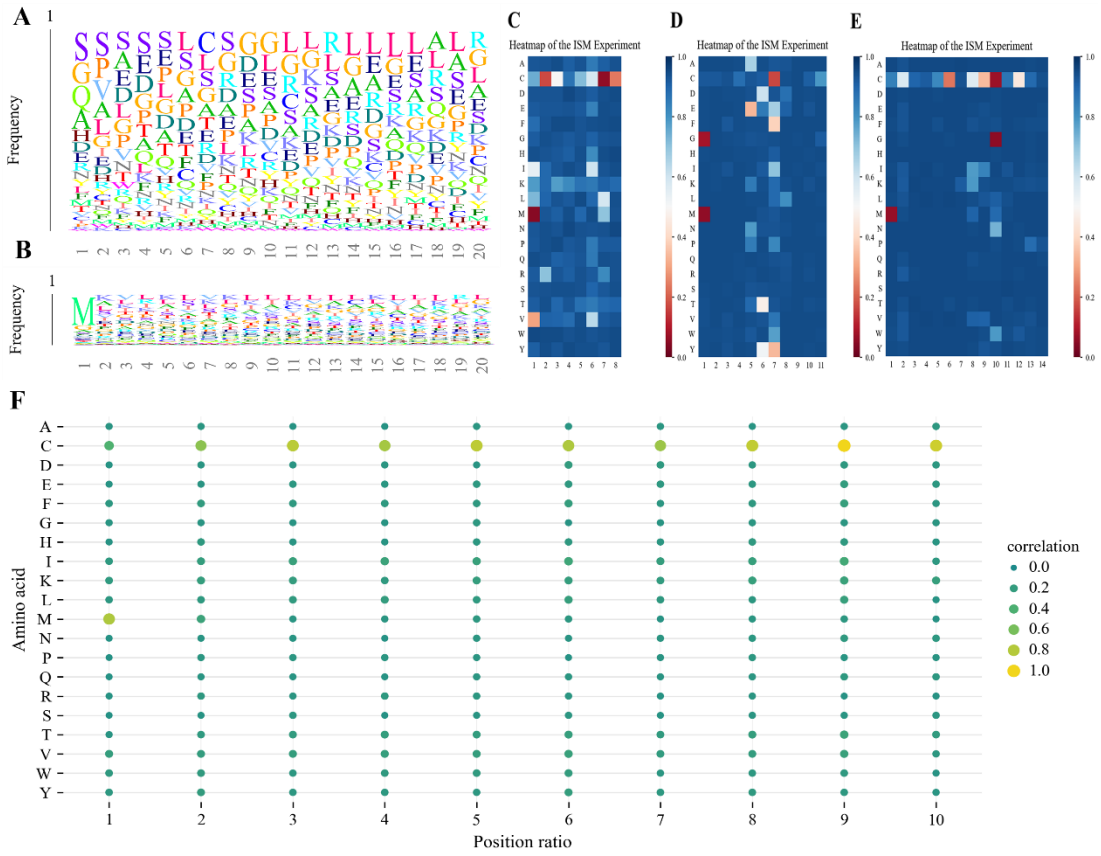


Figure 8. Mutation-scan analysis. (A, B) Position-specific amino acid preferences for neuropeptides and non-neuropeptides visualized by kpLogo. (C-E) ISM heatmaps of three representative peptides, with colors indicating normalized prediction scores from low (red) to high (blue). (F) Global ISM visualization illustrating amino acid contribution patterns across position-ratio intervals.